



University of Ghana · College of Basic and Applied Sciences
School of Engineering Sciences · Department of Computer Engineering

CPEN 208 — Project 1

Relational Database & Web Application
for the Computer Engineering Department

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Course: CPEN 208 — Introduction to Software Engineering

Semester: First Semester, 2025/2026

PostgreSQL

Next.js 14

TypeScript

Tailwind CSS

GitHub: <https://github.com/doxaodoi/project-1-2>

1. Overview

This project delivers a relational database for a software the Computer Engineering Department intends to build, together with a Next.js 14 web front end. The system supports five functionalities required by the brief:

#	Functionality	Implemented by
1	Student personal information	<code>people.students</code>
2	Student fees payments	<code>finance.fee_bills</code> , <code>finance.payments</code>
3	Course enrollment	<code>academic.enrollments</code>
4	Lecturers to course assignment	<code>academic.lecturer_course_assignment</code>
5	Lecturers to TA assignment	<code>academic.lecturer_ta_assignment</code>

A database function, `finance.get_outstanding_fees()` , computes each student's outstanding balance and returns it as a JSON array, as required.

1.1 Assumptions

- Database name: `coe_dept` . The design is split into four schemas for clarity: `people` , `academic` , `finance` and `auth` .
- The sample data is the real CPEN 208 class roster (70 students). Each student's email follows the University pattern `<studentID>@st.ug.edu.gh` .
- Supporting attributes not present in the roster (phone, date of birth) are generated sample values; gender is left unspecified rather than inferred from names.
- Each student is billed three fee items per semester (Tuition, Academic Facility User Fee, SRC & Other Dues) totalling **GH¢12,300.00**; payments vary so that outstanding balances are realistic.
- Every student has a login account (`auth.users`) with the demo password `Student@123` for grading.

2. Database Design

The database is organised into four schemas. The tables and their most important columns are shown below; primary keys are underlined in the SQL scripts and every relationship is enforced with a foreign key.

people

programs

program_id (PK), code, name, degree

students

student_id (PK), full_name, email, phone,
date_of_birth, level, program_id → programs

lecturers

lecturer_id (PK), staff_no, title, full_name, email,
academic_rank

teaching_assistants

ta_id (PK), full_name, email, student_id → students

academic

courses

course_id (PK), code, title, credit_hours, level,
semester

enrollments

enrollment_id (PK), student_id → students,
course_id → courses, academic_year, semester,
grade

lecturer_course_assignment

lecturer_id → lecturers, course_id → courses,
academic_year, semester

lecturer_ta_assignment

lecturer_id → lecturers, ta_id → teaching_assistants,
course_id → courses

finance

fee_bills

bill_id (PK), student_id → students, academic_year,
semester, description, amount_due

payments

payment_id (PK), student_id → students, amount,
paid_on, method, reference

auth

users

user_id (PK), student_id → students (unique), email
(unique), password_hash, role

2.1 Relationships (foreign keys)

Child table	Column	References
people.students	program_id	people.programs
people.teaching_assistants	student_id	people.students
academic.enrollments	student_id, course_id	people.students, academic.courses

academic.lecturer_course_assignment	lecturer_id, course_id	people.lecturers, academic.courses
academic.lecturer_ta_assignment	lecturer_id, ta_id, course_id	people.lecturers, people.teaching_assistants, academic.courses
finance.fee_bills	student_id	people.students
finance.payments	student_id	people.students
auth.users	student_id	people.students

2.2 Build scripts

The database is created by six ordered scripts in `PROJECT 1/database`, run in the pgAdmin Query Tool:

Script	Purpose
00_create_database.sql	Creates the <code>coe_dept</code> database
01_schemas.sql	Creates the four schemas
02_tables.sql	Creates all tables, keys and indexes
03_functions.sql	Creates the outstanding-fees functions
04_seed.sql	Inserts the 70-student class roster and all sample data
05_verify.sql	Row-count checks and a demonstration of the JSON function

3. The Outstanding-Fees Function

The required function aggregates each student's total bills and total payments and returns `billed - paid` as the outstanding amount, for every student, in a single JSON array.

```
CREATE OR REPLACE FUNCTION finance.get_outstanding_fees()
RETURNS JSON
LANGUAGE sql
STABLE
AS $$
    SELECT COALESCE(json_agg(row_to_json(t) ORDER BY t.full_name), '[]'::json)
    FROM (
        SELECT
            s.student_id, s.full_name, s.email,
            COALESCE(b.total_billed, 0) AS total_billed,
            COALESCE(p.total_paid, 0) AS total_paid,
            COALESCE(b.total_billed, 0) - COALESCE(p.total_paid, 0) AS outstanding
        FROM people.students s
        LEFT JOIN (SELECT student_id, SUM(amount_due) AS total_billed
                   FROM finance.fee_bills GROUP BY student_id) b ON b.student_id =
s.student_id
        LEFT JOIN (SELECT student_id, SUM(amount) AS total_paid
                   FROM finance.payments GROUP BY student_id) p ON p.student_id =
s.student_id
    ) t;
$$;
```

3.1 Sample output

Calling `SELECT finance.get_outstanding_fees();` returns a 70-element array. Three representative elements (a full payer, a partial payer, and the author's account):

```
[
  { "student_id": 22384451, "full_name": "Abu Neaquittae Golda",
    "total_billed": 12300.00, "total_paid": 12300.00, "outstanding": 0.00 },
  { "student_id": 22357814, "full_name": "Adzasa Stephen Yaw",
    "total_billed": 12300.00, "total_paid": 8000.00, "outstanding": 4300.00 },
  { "student_id": 22312110, "full_name": "David Kwame Odoi-Anim",
    "total_billed": 12300.00, "total_paid": 9000.00, "outstanding": 3300.00 }
]
```

3.2 Distribution across the class

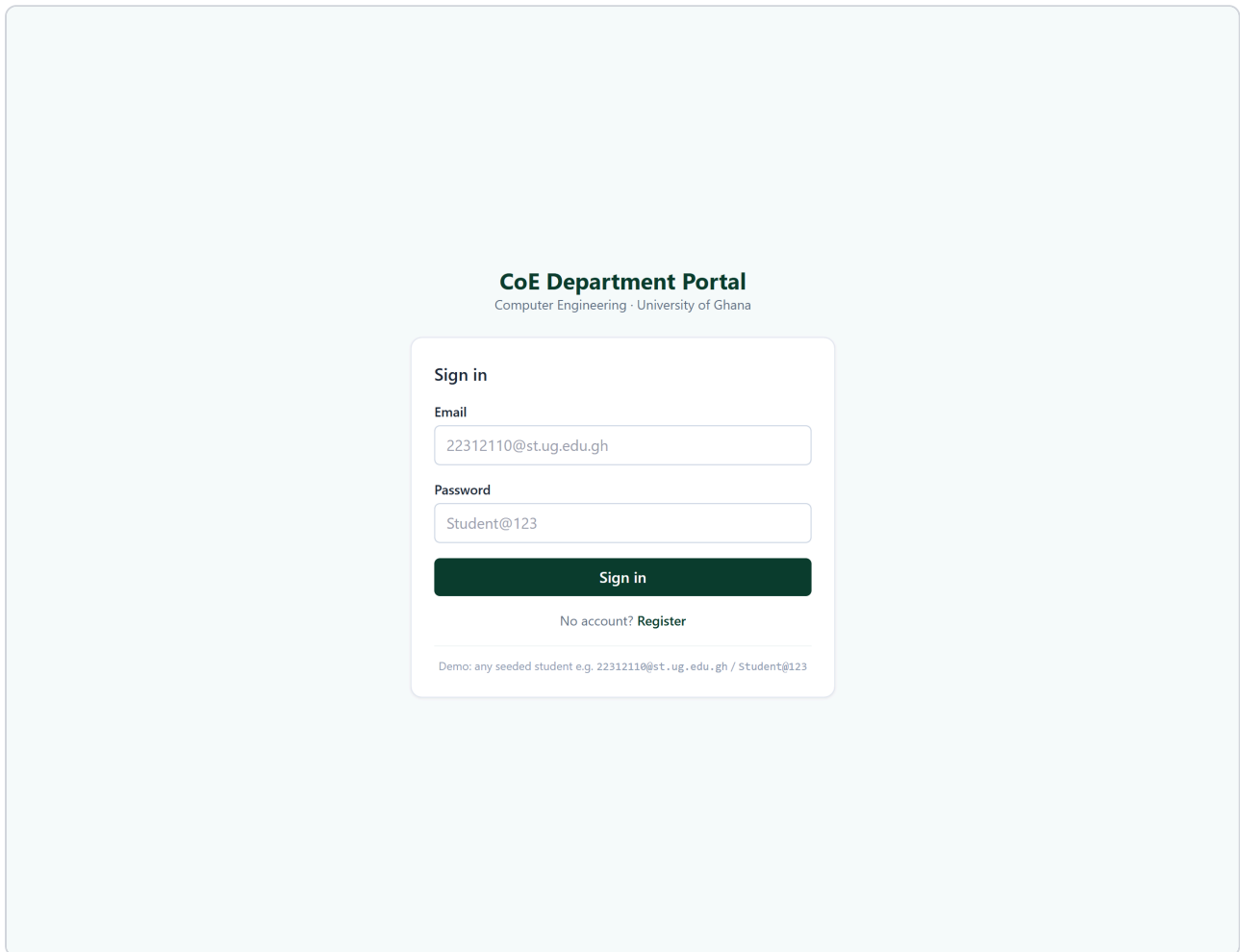
Outstanding (GH¢)	Number of students
0.00 (fully paid)	27

3,300.00	1
4,300.00	14
7,300.00	14
12,300.00 (no payment yet)	14

A scalar helper, `finance.get_student_outstanding(student_id)`, returns the balance for one student and is reused by the web application's dashboard.

4. Web Application (Next.js 14)

The front end is a Next.js 14 application (App Router, TypeScript, Tailwind CSS) with three pages — **login**, **register** and **dashboard**. It does not query the database directly; it consumes the Project 2 API. Authentication uses a JWT that the Next.js server stores in an HttpOnly cookie, and the `/dashboard` route is guarded by middleware.



The screenshot shows a web application titled "CoE Department Portal" with the subtitle "Computer Engineering · University of Ghana". The main content is a "Sign in" form. The form has two input fields: "Email" with the value "22312110@st.ug.edu.gh" and "Password" with the value "Student@123". Below the password field is a dark green "Sign in" button. Under the button, there is a link "No account? Register". At the bottom of the form, a small text line reads "Demo: any seeded student e.g. 22312110@st.ug.edu.gh / Student@123".

Login page — seeded students sign in with their student-ID email and the demo password.

Create your account

Computer Engineering · University of Ghana

Student ID

Full name

Email

Password

Register

Already have an account? [Sign in](#)

Registration page — creates a new student and login account.

Signed in as
David Kwame Odoi-Anim
Student ID 22312110
Email 22312110@st.ug.edu.gh
Programme BSc Computer Engineering
Level 200

Total billed
GH¢12,300.00

Total paid
GH¢9,000.00

Outstanding
GH¢3,300.00

Enrolled Courses

Code	Title	Credits	Lecturer	Grade
CPEN 202	Object Oriented Programming	3	Dr. Kwame Mensah	In progress
CPEN 204	Data Structures and Algorithms	3	Dr. Ama Boateng	In progress
CPEN 206	Linear Circuits	3	Dr. Yaw Owusu	In progress
CPEN 208	Introduction to Software Engineering	3	Prof. Elsie Adjei	In progress
CPEN 210	Digital Logic Design	3	Dr. Efua Danso	In progress
MATH 223	Multivariable Calculus	3	Mr. Kofi Antwi	In progress

Payment History

Date	Amount	Method	Reference
2025-06-12	GH¢6,000.00	BANK	PAY-22312110-1
2025-07-14	GH¢3,000.00	MOMO	PAY-22312110-2

Dashboard for student 22312110 — profile, live fee summary (billed / paid / outstanding from the database function), enrolled courses with assigned lecturers, and payment history.

5. How to Run

1. **Database:** in pgAdmin, run `00_create_database.sql` against the `postgres` database, then run `01 – 05` against `coe_dept` . (Or restore `coe_dept_backup.backup` .)
2. **API (Project 2):** `cd "PROJECT 2/api"` , set `DB_PASSWORD` , then `mvn spring-boot:run` (starts on port 8080).
3. **Front end:** `cd "PROJECT 1/web"` , copy `.env.example` to `.env.local` , then `npm install && npm run dev` and open `http://localhost:3000` .
4. Sign in with `22312110@st.ug.edu.gh` / `Student@123` .

Deliverables in this folder: the SQL scripts and database backup are in `PROJECT 1/database` ; the front-end source is in `PROJECT 1/web` ; all source code is on GitHub at <https://github.com/doxaodoi/project-1-2> .